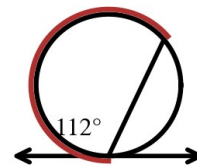


Speak Like A Mathematician:

circumscribed angle:

the big idea:



Vertex Location Rules

Vertex ON the circle (tangent-chord): angle = _____ of the intercepted arc.

Vertex INSIDE the circle (two chords): angle = half the _____ of the two vertical arcs.

Vertex OUTSIDE the circle (secants / tangents): angle = half the _____ of the far and near arcs.

Circumscribed angle = 180° – the _____ angle that opens to the same arc.

Example 1 - Tangent and Chord

In the figure at the top right, the tangent-chord angle measures 112° .

- Find the measure of the marked (major) intercepted arc.
- Find the other arc and the angle on the other side of the chord.

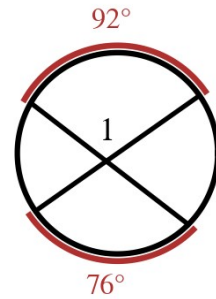
Example 2 - Vertex Inside

Two chords cross in the circle at the right. The vertical arcs measure 92° and 76° .

- a) Find $m\angle 1$.
- b) What is the angle NEXT to $\angle 1$?

Example 3 - Inside, Solving Backward

- a) Two chords cross forming a 71° angle. One vertical arc is 58° . Find the other arc x .



Example 4 - Vertex Outside: Two Secants

In the top figure, the far arc is 158° and the near arc is 62° .

- a) Find $m\angle P$.

Example 5 - Two Tangents

In the middle figure, the two arcs measure 236° and 124° .

- a) Find $m\angle Q$.
- b) Check with the circumscribed-angle shortcut.

Example 6 - Outside, Solving Backward

In the bottom figure, the exterior angle is 34° and the near arc is 52° .

- a) Find the far arc x .

